

A cluster-based decision support system for estimating earthquake damage and casualties

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This paper describes a Decision Support System for Disaster Management (DSS-DM) to aid operational and strategic planning and policymaking for disaster mitigation and preparedness in a less-developed infrastructural context. Such contexts require a more flexible and robust system for fast prediction of damage and losses. The proposed system is specifically designed for earthquake scenarios, estimating the extent of human losses and injuries, as well as the need for temporary shelters. The DSS-DM uses a scenario approach to calculate the aforementioned parameters at the district and sub-district level at different earthquake intensities. The following system modules have been created: clusters (buildings) with respect to use; buildings with respect to construction typology; and estimations of damage to clusters, human losses and injuries, and the need for shelters. The paper not only examines the components of the DSS-DM, but also looks at its application in Besiktas municipality in the city of Istanbul, Turkey.

Keywords: cluster approach, decision support system, disaster management, earthquakes, fast prediction, less-developed contexts

Introduction

The Kocaeli earthquake on 17 August 1999 and the Düzce earthquake on 12 November 1999 claimed the lives of more than 18,000 people, destroyed 16,400 buildings and caused massive disruption to the economy of western Turkey (Erdik, 2001). The Kocaeli earthquake is considered to be the largest event to devastate a modern, industrialised area since the 1923 earthquake in Tokyo, Japan. This earthquake tragically illustrated the country's inability to cope with the results of poorly controlled urban development and rapid population growth in disaster-prone regions. The price of inadequate preparedness was the loss of thousands of lives, the displacement of thousands of people from their homes, and economic costs of more than USD 20 billion (Harrald et al., 2002).

Just to the northwest of the hardest-hit region is Istanbul, a rapidly growing city that has suffered heavy damage due to earthquakes 12 times in the past 15 centuries. The Kocaeli earthquake caused significant building damage in a western suburb of Istanbul, Avcılar, despite the fact that it lies approximately 100 kilometres from the source zone. A recent study reports the probability of strong tremors in Istanbul—an urban centre of ten million people—as 62 plus or minus 15% over the next 30 years and 32 plus or minus 12% over the next decade (Parsons et al., 2000).

The threat of a devastating earthquake has pointed up the urgent need to develop a comprehensive disaster management system for Istanbul. Some recent projects have been completed, such as the Istanbul Metropolitan Municipality Earthquake Master Plan, 2003. With regard to the latter, many aspects of a disaster management system have been finalised, such as the relocation of residential areas, structural retrofitting, administrative restructuring and supporting financial/insurance tools. Such measures require contributions from various fields of expertise and may take considerable time to draw up and implement, thus requiring long-term planning. Long-term planning and policymaking have become important in 'risk societies' (Kleiboer, 1997), where protection against negative contingencies is underlined.

Disaster preparedness has been stressed in countries with less-developed disaster management systems following a major disaster. The tendency within these nations is to deal with the disaster after it happens. The decision to focus more on pre-disaster stages is itself a fundamental change in disaster management in these contexts. Existing systems employed in developed states, such as Hazards-US (HAZUS) (FEMA, 1999), though, are not always directly applicable and useful in nations where key social and economic indicators reveal differences. Such countries lack the necessary infrastructure to adopt in full the well-integrated systems developed in nations where the context can be vastly different in terms of availability and accuracy of data, the number of experienced personnel and the level of funds. As will be shown later in this paper, the HAZUS methodology relies heavily on high precision in relation to data collection and treatment, and advanced engineering analysis. Such requirements are considered inappropriate for contexts where the operational infrastructure renders highly developed and precise methodology unusable. It is thus rational to develop context-dependent, low-cost, flexible systems in these areas, as opposed to directly adopting highly sophisticated systems that do not necessarily satisfy infrastructural requirements. It has been emphasised recently that emergency response decision systems need to be revised to become more flexible and creative (Mendonça et al., 2001). This paper proposes such a model, tailor-made for less-developed infrastructural contexts.

One of the main issues in developing such a system is estimation of casualties and their distribution across the territory, which forms part of the disaster-management risk assessment (Terry, 2001). The expected number of fatalities and injuries is a function of the expected damage to buildings exposed to the earthquake. Thus, in a particular earthquake situation, the vulnerability of buildings becomes the most important variable in determining the extent of casualties. However, such estimations may be highly unreliable due to the impossibility of precisely assessing the earthquake parameters (time, place and magnitude, for instance) beforehand. The difficulty in obtaining reliable estimates is increased by incomplete knowledge of how parameters like local soil conditions, construction type, structural systems and construction quality interact.

Although scientists have been trying to improve predictions about earthquakes, uncertainty remains about how various factors interact and about their relative importance (Kunreuther, 2000). Earthquake-related uncertainty can be reduced in part through a detailed study of the building stock and local soil conditions. Conducting such a

study in Istanbul, where there are more than one million buildings, would require vast amount of resources—in terms of time, money and qualified personnel. Hence, pilot projects are being planned that focus primarily on geologically higher risk areas, such as Zeytinburnu.

More developed software systems that are used as decision support tools for safety management (Lazzari and Salvaneschi, 2000) and HAZUS tend to rely heavily on Geographic Information Systems (GIS) data. The expertise and precision required for such data makes the decision support tool much more complicated and advanced for end users. Another example of a more complex tool is the Early Post-Earthquake Damage Assessment Tool (EPEDAT), developed at the Central Office for Emergency Services in California (Eguchi et al., 1997). The authors admit that modifications (such as smaller units of analysis) have to be made to its configuration to meet the needs of various organisations, since it was designed for macro-level estimation. The imminent threat of earthquakes, though, necessitates the rapid design of a predictive model that uses available and easily accessible data to enable fast scenario analysis. Furthermore, threat assessment needs to be performed both before and during the disaster (Perry and Lindell, 2003), highlighting the importance of a fast and flexible model.

It has been suggested in studies conducted after large-scale disasters, such as the earthquake in Gujarat, India, on 26 January 2001, that information, communication and loss estimation are of great importance in the post-disaster period (Sharma, 2001). Indeed, in order to respond effectively to disasters, it is necessary to have both exact and reliable information on the situation and well-trained personnel at all levels of management (from administration to search-and-rescue entities to technical support groups). When an earthquake strikes, it is extremely difficult to obtain immediate and reliable information due to communication problems.

Thus, a scenario approach can be utilised in disaster management programmes (Drabek, 1986; Foster, 1980). Scenarios are accepted as important disaster management tools (Kleiboer, 1997), increasing disaster preparedness by providing a range of possible future states. This approach allows researchers to model the situation in advance. The information generated from various scenarios forms a basis for the development of further planning activities, including priority training programmes for relief personnel and the general population, with the ultimate aim of decreasing the number of fatalities and injuries. Moreover, scenarios for the hypothetical construction of future events are useful in developing skills like time management, cognitive mapping, mediation and team management, as well as the ability to make decisions under stress (Alexander, 2000). Scenarios, therefore, are important heuristic tools for policy planning and strategy development for agencies involved in disaster-related activities.

In disaster management, preparedness and mitigation strategies involve providing warning, establishing contingency plans, developing capacity for emergency response and reducing vulnerability and risk on a long-term and permanent basis (Nateghi-A, 2000). According to Wallace and De Balogh (1985), decision support system technology is considered to be a useful tool in the preparedness, mitigation, response and recovery stages of a disaster event. The same authors point to the non-routine nature of disaster

management activities, which may lead to difficulties with regard to the ‘structurability’ (Wallace and De Balogh, 1985, p. 136) of a disaster decision process. In such cases, the model needs to be flexible to allow for ‘what if’ assessments. During earthquake preparedness and mitigation efforts, risk assessment to estimate building damage and human casualties needs to precede other planning activities. The Decision Support System for Disaster Management (DSS-DM) has been designed for this very purpose. The DSS-DM evaluates human losses and injuries as well as the need for temporary shelters at both the district and sub-district level of the city of Istanbul. It comprises several stages. First, the system introduces the concept of a cluster in order to collect key input information: on population and the construction typology of buildings. After earthquake scenarios have been designed with respect to their possible impact, earthquake resistance offered by different construction typologies is fed in to the system. Depending on the level of damage sustained by buildings of each construction typology, one can obtain an estimate of human losses and injuries. Finally, the DSS-DM is extended to include a model for the need for shelters.

The remainder of this paper is organised as follows: Section 2 describes the main typologies used in the DSS-DM. It presents the concept of a cluster and develops the construction typology both for the use of clusters and building type. Section 3 outlines the general structure of the decision support system. Section 4 sets out the hardware and software requirements of the DSS-DM. Section 5 describes the application of the DSS-DM in different earthquake scenarios in Besiktas municipality, Istanbul, to estimate human losses and injuries, as well as the need for shelters.

Typology of information used in the DSS-DM

Loss prediction models are designed to calculate physical damage to a structure and infrastructure. This damage can then be used to gauge the resulting human impacts and service delivery needs (Perkins et al., 2000). In line with the discussion above, the model developed in this paper is flexible in terms of redefining and/or adding categories. The DSS-DM includes concepts and categories accepted as valid following feedback from experts in search-and-rescue teams, civil engineers and public administrators. Lack of a detailed building stock analysis for the city of Istanbul that might be used in damage predictions led to the development of the cluster concept. Data on two major inputs—population and building stock—are gathered by employing the cluster as a unit of information in the DSS-DM.

Districts and sub-districts

The smallest official administrative unit in Turkish cities is the *district*. Decisions on district boundaries are usually made on a historical basis—most of the current districts were villages before they were incorporated into the administrative boundaries of the city. Each district has a population varying from a few hundred to several thousand people. The district administrator, in consultation with the council of elders, governs

it. The district administrator is elected for a term of five years. He/she represents the central administration and supervises the planning and operation of district projects and services.

The DSS-DM incorporates the term *sub-district*, which refers to natural divisions in a district. During the data collection phase of this study, the district administrator, based on his/her knowledge of the district, defined the sub-districts in a district.

Clusters

The unit of information in the DSS-DM is a *cluster*. A cluster consists of several buildings of the same type and utilised for the same purpose (for example, all residential masonry buildings of between one and three stories in a sub-district). Alternatively, it consists of a single building that many people use simultaneously at a given time (for instance, big trade centres that might accommodate a few thousand people at particular hours of the day, or an elementary school with several hundred children during the day). In a practical sense, clusters are conceptually created population concentration centres to facilitate data collection on major inputs to be used in the DSS-DM.

The cluster types used in the DSS-DM are as follows: residential; elementary schools and pre-schools; high schools; universities; hypermarkets; trade centres; recreational sites; business centres; general offices; warehouses; factories; and military buildings. Information is collected on the construction typology and the number of people residing in a specific cluster during the day and night for all clusters at the sub-district level (see Appendix 1). The DSS-DM uses aggregate cluster information at the sub-district level.

Construction typology

Building-specific fatality estimations based on structural damage can be made using detailed analyses (Yeo and Cornell, 2003) and are accepted as important for disaster management. However, the model presented here emphasises fast prediction and the concept of cluster rather than building-specific detailed analyses. Three major factors, namely, construction typology, number of stories, and construction year, are employed to gauge the earthquake performance of a building (Applied Technology Council, 1992; Applied Technology Council, 1985; Doyduk, 2003; Smyth et al., 2004). All categorisations in this section are expert assessments (by civil engineers, municipality administrators and engineering associations) based on the current building stock in Istanbul. The methodology described herein can be adapted to any country provided that the parameters used are obtained from that particular region's construction practices and relevant data.

In assessing the effects of an earthquake, the distinction between buildings with different structural systems must be taken into account. In this study, the structural systems are classified as follows:

- masonry;
- reinforced concrete;

- wooden masonry composite; and
- steel frame.

Another parameter that must be considered is the number of stories. Buildings are grouped as follows:

- between one and three stories;
- between four and five stories; and
- six or more stories.

For some structural systems, though, storey classification is modified to allow for more meaningful categories for that specific building type. For example, in the case of wooden masonry composite buildings, categories are classified as 'five stories and less' and 'six stories and more'.

The construction year is also a very significant variable. Both the technology and building codes available at the project development and construction stage are important in determining the earthquake resistance of a building. New building codes were introduced in 1975 and 1998 in Turkey, improving the resistance of buildings by incorporating new research results and bringing in tighter standards. However, lack of enforcement of codes and lack of control of construction practices resulted in the emergence of buildings that do not comply with requirements. According to expert interviews (Applied Technology Council, 1992; Erdik and Aydinoglu, 2002; Smyth et al., 2004) and surveys (Architectural Institute of Japan, 2001) conducted after the 1999 Kocaeli earthquake, the time intervals for the construction year are to be categorised as follows:

- before 1940;
- 1940–49;
- 1950–74;
- 1975–84;
- 1985–97; and
- 1998 and after.

It is worth noting that year categorisations might be collapsed for some structural systems if there is no difference in terms of vulnerability. For example, steel frame buildings are treated as two categories: 'before 1940' and '1940 and after'.

In the HAZUS earthquake loss methodology, the structural damage estimation is standardised and an analytical method describing vulnerability curves in terms of spectral displacements is used. Technically, the method employs inelastic drifts of buildings for various damage states. The drift demand of a typical building is estimated using non-linear static procedures, which are based on the capacity curve of the given building. The inelastic spectral displacement demand is estimated based on the capacity curve. The analytical approach used in the HAZUS methodology is cumbersome and requires expertise both in advanced engineering analysis and data collection and treatment. The model presented in this paper is relevant for cases where data are incomplete and where fast loss estimation can be traded for higher precision.

Data collection methodology

Data are collected at the district and sub-district level from district administrators using a questionnaire on population and the construction typology of clusters. As the exact time of an earthquake cannot be forecast, the distribution of the population in various clusters during the day and night needs to be taken into account. The questionnaire includes questions on the daytime and the night-time populations of a cluster (see Appendix 1).

The pilot district where the DSS-DM was applied was Rumeli Hisarüstü in the northern part of Istanbul that flanks Europe. It was selected as a convenient site because of its proximity to Bogaziçi University, where this research was done. The district administrator devoted a generous amount of time to helping to supply cluster pertinent data on the sub-districts. The meaningfulness and comprehensiveness of the categories and typologies were validated during the pilot study.

In the process of preparing the questionnaire, the main concern was availability and accessibility of data. The mayor invited district administrators in Besiktas municipality to a meeting to provide feedback on the questions. It was observed that those administrators who have been working in their district for more than two years could easily complete the questionnaire. In a relatively small number of cases, members of the council of elders provided data for some of the questions.

The structure of the DSS-DM

In using a decision support system, decision-makers can explore alternatives and evaluate their consequences. Such a system can produce several iterations that can be used for disaster mitigation and response planning. Examples of such decision support systems can be found in Berke and Stubbs (1989) (hurricane mitigation planning) and in Stojanovski and Dong (1994) (earthquake casualty estimation using an event tree model).

Figure 1 depicts the structure of the DSS-DM developed in this paper. The system is explained in detail below.

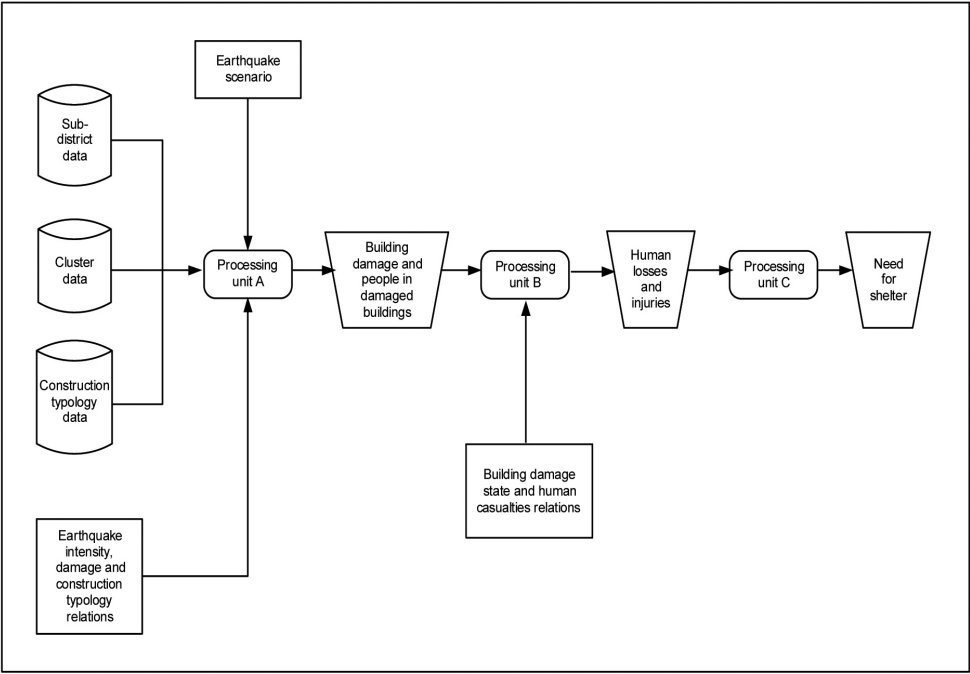
Earthquake scenarios

In the DSS-DM, different earthquake scenarios are developed using the intensity of a possible earthquake as a variable that takes values of *VII*, *VIII* and *IX* as defined by European Macroseismic Scale 98 (EMS). The DSS-DM allows for the specification of one of these intensities for each district. Therefore, with n districts in a city, one might construct 3^n scenarios. To ensure that scenarios are realistic, the input of seismologists is crucial at this stage. The choice of the time of day (daytime/night-time) completes the DSS-DM scenario construction.

Estimation of building damage

If Q is a set of earthquake intensities, where $Q = \{VII, VIII, IX\}$, and C is a set of construction typologies, where $C = \{c_1, \dots, c_m\}$, then the estimation of building damage is

Figure 1 DSS-DM conceptual model



given in the functional form $Q \times C \rightarrow D_1 \times D_2 \times D_3 \times D_4$, where $D_i = [0, 100]$ for $i=1,2,3,4$. Naturally, $\sum D_i = 100$.

The values D_1 , D_2 , D_3 and D_4 define the percentages of collapsed/heavy damage, moderate damage, slight damage and no damage, respectively. The damage states adopted for this model differ from those employed in the HAZUS software. This model combines the extensive damage and complete collapse categories of HAZUS into one single category entitled ‘collapsed/heavy damage’.

In the DSS-DM, the building damage percentages are expert estimations. The latter are very powerful instruments to solve weakly structured problems (see, for example, Bonczek et al., 1983; Ericsson, 1996; Larichev, 2000). Similar expert assessments or predictions based on statistical knowledge can be found in disaster-related studies. For instance, the relationship between the earthquake vulnerability of general mid-rise reinforced concrete frame buildings and earthquake intensity in Turkey is set out in Erdik, 2001.

Table 1 presents estimated damage percentages for a particular building type (reinforced concrete buildings with between four and five stories and constructed in 1975–84) used in the DSS-DM. For example, an earthquake intensity of IX is expected to lead to 15% of the buildings of this construction typology in a cluster collapsing or suffering heavy damage.

The Processing Unit A of the DSS-DM (see Figure 1) uses building damage estimates and the number of people in each cluster to determine the number of people present

in buildings with different levels of damage at the sub-district level. These results are stored in a building damage database (see Table 3 in Appendix 2). Processing Unit A also produces estimates of the number of people in the buildings with different levels of damage listed by construction typologies at the sub-district level (see Table 4 in Appendix 2).

Estimation of human losses and injuries

The underlying idea of this module is that the extent of human losses and injuries in a particular earthquake scenario depends on the construction typology and the damage state of the building. For instance, for a given earthquake intensity and a fixed distribution of building damage, the percentage of seriously injured people in a wooden/masonry mixed building with between four and five stories and built in 1975–84 differs from the percentage of seriously injured people in a reinforced concrete building with between four and five stories and built in 1975–84.

In order to estimate human losses and injuries, the model assumes c to be the construction typology of a cluster. Then we define a function λ that evaluates the percentage of deaths (p_1), seriously injured (p_2), moderately injured (p_3), slightly injured (p_4) and unaffected people (p_5) in that cluster for each damage level d . In other words, λ is a function: $C \times D \rightarrow P_1 \times P_2 \times P_3 \times P_4 \times P_5$, where $P_i = [0, 100]$ for $i=1,2,3,4,5$. Naturally, $P_1 + P_2 + P_3 + P_4 + P_5 = 100$.

As Erdik and Aydınoglu (2002) suggest, casualty rates derived for other regions of the world cannot be directly transferred to Turkey. In the 1992 earthquake in Erzincan, Turkey, for instance, the observed damage rate for four-to-six storey reinforced concrete buildings was four times higher than that experienced in the 1995 earthquake in Hanshin, Japan, and 12 times higher than that experienced in the 1995 earthquake in Northridge, US (Erdik, 1999). Therefore, context-dependent casualty rates need to be developed. In the DSS-DM, λ is defined on the basis of statistical information obtained in previous earthquakes. It has also been suggested (Yeo and Cornell, 2003) that individual behaviour during an earthquake inside a building can have an important impact on fatality estimation. This study suggests that there is human tendency to move

Table 1 Damage estimations for construction typologies (percentage)*

Earthquake intensity	Damage state			
	Collapsed/ heavy damage	Moderate damage	Slight damage	No damage
VII	0	0	3	97
VIII	5	10	25	60
IX	15	15	50	20

* Reinforced concrete buildings with between four and five stories and constructed in 1975–84.

towards exits; thus building more robust stair cores where the majority of the occupants will be situated, affects fatalities. However, such assumptions and information are more useful inputs for reinforcement efforts or new buildings.

Another point that needs to be emphasised is the insufficient distribution of information and inadequate organisation efforts with respect to preparedness and mitigation in some communities vulnerable to seismic risk, as with the case study in this paper. Research carried out following the Osmaniye earthquake of 25 June 2001 indicates that, although behavioural coping information was mentioned by participants as one of the most useful types of information, it was not used significantly during the earthquake (CENDIM, July 2001). The same research reports that participants accorded the highest importance to structural safety issues to reduce damage. It has also been suggested (Inelmen et al., 2004) that participation in earthquake preparedness efforts in the Turkish context tends to be low due to various cultural and sociological factors. A common point in these studies, though, is that prior earthquake training and education programmes need to be established to ensure a stronger preparedness phase, while accepting that there is not a systematic attempt to organise and evaluate such programmes in Turkey, except for a few discrete attempts, such as the KOERI–IAHEP.⁶

These results highlight the context-sensitive nature of preparedness measures and the need to evaluate developed models for their direct applicability in such contexts. The informed human behaviour, which varies in different contexts, is embedded in the casualty rates. These rates implicitly endorse the effect of this parameter and are defined on the basis of previous local earthquake statistical information. Values of λ for the construction type ‘reinforced concrete building with between four and five stories and constructed in 1975–84’ for different levels of building damage are provided in Table 2 by way of example.

Using the function λ and data on the number of people in a cluster at the particular time of day that the earthquake strikes, Processing Unit B estimates the number of deaths, seriously injured, moderately injured and slightly injured people in each cluster. The estimates for the sub-districts and districts are obtained through summation of cluster casualties (see Table 5 in Appendix 2).

Table 2 Casualty estimations for construction typologies (percentage)*

Damage state	Casualty state			
	Deaths	Seriously injured	Moderately injured	Slightly injured
Collapsed/ heavy damage	6	8	11.5	30
Moderate damage	0	0	0.02	0.2
Slight damage	0	0	0.005	0.05

* Reinforced concrete buildings with between four and five storeys and constructed in 1975–84.

Need for shelters

The underlying assumptions of the model estimating the need for shelters are as follows:

- only people in residential clusters need shelters;
- only slightly injured and unaffected people need shelters;
- people will only return to slightly damaged or undamaged houses after the earthquake;
- even if their homes are undamaged, the willingness of people to return to them depends on the earthquake intensity; and
- the willingness of people to return to their homes depends on the construction type of the building.

The last two assumptions in particular have a psychological basis. Indeed, after a high-intensity earthquake, people prefer to stay outside (at least for a few days), even if their own house has not been damaged. The explanation for this may be the expectation of aftershocks and additional earthquakes. The need for shelters is also affected by a number of other factors, such as weather conditions, the presence of relatives either in hospitals or in safe houses, the quality of water supplies and the availability of out-of-area transport. These variables, though, are not included in this study, although they are valuable points for further research.

In modelling the need for shelters, let δ_i be the casualty rate in residential cluster i , which is defined as:

$$\delta_i = \frac{(n_i^d + n_i^h + n_i^m)}{n_i}$$

In this equation, n_i^d is the number of deaths, n_i^h is the number of seriously injured people, n_i^m is the number of moderately injured people and n_i is the number of people in cluster i . Obviously, δ_i varies between zero and one, and depends on the construction typology.

Let n_i^s be the sum of slightly injured and unaffected people in cluster i , that is, $n_i^s = n_i - (n_i^d + n_i^h + n_i^m)$, and let n_i^{sh} be the number of people in cluster i who need temporary shelters.

Finally, if q represents the intensity of the earthquake, then:

$$n_i^{sh} = \begin{cases} n_i^s, & \text{if } q = IX; \\ 0.3n_i^s, & \text{if } q = VIII \text{ and } \delta_i \leq 0.01; \\ 0.9n_i^s, & \text{if } q = VIII \text{ and } \delta_i > 0.01; \\ 0, & \text{if } q = VII. \end{cases}$$

In other words, if the earthquake intensity is IX, people in residential areas, regardless of the condition of their house, will prefer to stay outside. If the earthquake intensity is VIII, and the casualty rate of the cluster is less than or equal to 0.01, the majority

of people (70%) will return home. For the same earthquake intensity, if the casualty rate is greater than 0.01, only ten per cent of people will return home. Finally, if q is equal to *VII*, there is no need for temporary shelters. It is worth noting that rescue teams in Turkey provided the parameters of the function defining n_i^{sh} during personal interviews. Developing a more comprehensive needs assessment function is an important and valuable future research task, requiring a more detailed study of numerous variables.

Lastly, the number of people in a sub-district who will need shelter is defined by the summation of all residential clusters in a sub-district. That is:

$$n^{sh} = \sum_i n_i^{sh},$$

where i is the index characterising cluster i .

Table 6 in Appendix 2 presents estimates of the need for shelters in all sub-districts of Bebek district following an earthquake of intensity *VIII* that occurred during the day.

Special features of the DSS-DM

The system is constructed to allow for easy revision of the structure of input information and the basic structure of databases on construction typology and clusters. Naturally, all operational rules used in Processing Units A, B and C can be easily edited.

The DSS-DM has a set of editors allowing changes in:

- the structure of clusters;
- the structure of construction types;
- the structure of districts and sub-districts;
- the tables that relate construction typology and building damage;
- the tables that relate building damage and human losses and injuries; and
- the parameters of the model that define relations between building damage, earthquake scenarios and the need for shelters.

Another specific feature of the system is the use of fast algorithms. In an actual earthquake situation, a fast-running decision support system is essential when information is constantly changing. System results should be acquired in seconds, in order to respond effectively to relief work needs during the disaster.

Hardware and software requirements

The DSS-DM is a PC-based application software that was developed using Delphi 4.0 and Microsoft Access. The system’s hardware and software requirements are very modest—the minimum requirements are a Pentium 100 processor, 32 megabytes (MB) of main memory, 1.2 gigabytes (GB) of hard disk space and a Windows 98 or NT 4.0 operating system—compared to those of HAZUS-99.

During the software design phase, several design issues were considered:

- **User friendliness** Since the software will be used in critical operations, one of the main design goals was to make the software as user friendly as possible (to reduce errors). The DSS-DM is a software product that utilises Windows' user-friendly graphical interface components, such as tree objects, text boxes, labels, buttons and forms. Every individual familiar with Windows' applications can use it. Consequently, training time is greatly reduced. Additionally, an authorised user can enter, modify and delete particular information while preserving the consistency of the overall data.
- **Performance** Another design goal was to develop a system that can be deployed effortlessly and can be used under demanding time constraints. Managers using the software must be able either to identify easily the necessary precautions to take against an impending disaster or to make the best decision regarding relief efforts after a probable disaster in a timely manner. Because of the design, the DSS-DM software does not require large amounts of hardware and software resources. It can be dispatched to the field with an easily obtainable and relatively cheap configuration. The parts of the system are designed and implemented in the most efficient way in order to achieve the minimum possible response time from the DSS-DM. Likewise, the tables and queries in the database application were designed to minimise disk space usage and to maximise system performance.
- **Maintenance** Since changes to the software are inevitable (to increase its lifespan), the design considered maintainability. This was done by adhering to international software engineering standards.
- **Security** Ensuring that the stored information is safe and secure before and even after the probable disaster was another design goal. The DSS-DM offers protection against unauthorised access. A person requires a username and a password. Different user levels with different capabilities can also be defined.

Application of the DSS-DM in Besiktas municipality

Besiktas municipality, comprising both residential and business units, accommodates over 200,000 people. It has 12,989 buildings, 78,131 residential units and 10,017 business units. A number of universities and other educational institutions are located in the area. These institutions, together with big business centres, significantly increase the number of people among the daytime population.

Data collection

The mayor of Besiktas municipality asked district administrators to attend a meeting (in May 2000). The researchers explained in detail the structure of the data collection forms and the information expected from district administrators. Following an interactive question-and-answer session, the administrators were asked to return completed data forms within a week. Once the forms had been returned, the researchers examined them.

In order to check the validity of the information and to fill in gaps, the mayor arranged a second meeting (in June 2000). The data forms were completed after two or three rounds with the informants. On average, the meetings lasted 90 minutes, and all of the districts of Besiktas municipality were represented. In total, the maximum amount of time spent by district administrators on completing the forms was around 4.5 hours. However, the majority of administrators completed the data forms in approximately 1.5 hours.

Input data

Besiktas municipality has 22 districts. Bebek district, located in the northern part of Istanbul along the Bosphorus Strait, has five sub-districts. Data collected for the sub-district of Büyük Bebek in this district are presented in Appendix 1. This sub-district consists of six groupings of residential clusters (A, B, C, D, E and F), with buildings of different construction types. The number of people in these clusters during daytime is 175, 600, 80, 80, 150 and 200, respectively. There is one pre-school, one elementary school and Bogaziçi University (southern campus).

Output data

Appendix 2 contains DSS-DM output data for Büyük Bebek. The results for the other four sub-districts of Bebek district—Asiyan, Çamlı Bahçe, Cevdet Pasa and Küçük Bebek—are also included in Appendix 2.

Table 3 in Appendix 2 outlines the distribution of people in damaged (building) clusters; Table 4 notes the distribution of people in damaged buildings by typology of buildings; Table 5 details the casualty distribution in clusters and sub-districts; and Table 6 shows the number of people who need shelters (by sub-districts).

Given the information in Table 5, a death rate estimate for Bebek district can be calculated: 0.15%. This figure is in line with previous estimates and expert estimations and is consistent with data on Turkish earthquakes. Two studies provide macro-level fatality estimates. Tezcan and Gürsoy (2002) assume a loss of life percentage of 0.4 for Istanbul in their study of the building rehabilitation programme. Based on previous earthquake data in other areas of Turkey, Erdik (1999) offers the same estimate for Istanbul and further suggests a 0.2% fatality rate for Besiktas municipality. The model presented here can be said to produce micro-level results within limits accepted as valid in studies on the Turkish context.

Conclusion

Fabricated or natural disasters have become quite central to the everyday life of individuals and organisations, especially in recent years. The need to develop disaster preparedness systems has thus increased at the organisational and national level (Bui and Sankaran, 2001). The complementary objective of designing the DSS-DM was to provide a risk assessment of the city under investigation and to develop a disaster response capacity. As noted by various studies in this area, though, sophisticated and

complicated systems may not be directly applicable in less-developed contexts where the availability of data, trained personnel and resources places constraints on the implementation of such systems. The DSS-DM is a decision support system that aims to overcome such problems and to offer reliable predictions. Such a system is required in a country like Turkey, where the operational constraints and the risk of disaster are great. The DSS-DM addresses the following disaster management issues:

- the distribution of people in clusters (with different levels of damage) at the sub-district level;
- the distribution of people in buildings with different construction typologies at the sub-district level;
- the number of dead, seriously injured, moderately injured and slightly injured people in each cluster at the sub-district level; and
- the need for shelters at the sub-district level.

It differs from previous systems in that it is capable of producing micro-level (sub-district) estimates that enhance the quality of disaster management planning. The robustness of the system allows it to cope with resource and expertise scarcities and incomplete, inaccurate or missing data, while taking loss prediction to the sub-district level. The system has several other important advantages. First, initial data are easily accessible—data are collected from district administrators using a questionnaire. Second, data collection does not require that qualified personnel work with district administrators. In applying the DSS-DM in Besiktas municipality, university students were asked to check and review the data collected from administrators. It is estimated that initial data can be collected within two weeks for the city of Istanbul, if the governor's office and the city and municipal councils provide necessary support.

Another major advantage pertains to the flexibility of the information structure used in the DSS-DM. One can easily alter the district–sub-district division and edit all modules, allowing one to obtain new information from initial data, such as expert knowledge, statistical data and the parameters of models. As has been mentioned, the effects of additional parameters, including the impact of informed human behaviour on casualty estimations, can be incorporated into the model. The structure of the main informational units, clusters and construction types can also be modified. For example, the system can easily be expanded by incorporating new decision support system modules that will provide estimates on the required volume of the different blood groups—an initial planning tool for the health services.

The DSS-DM can be further improved and expanded by incorporating additional local information. For instance, for each district and sub-district level in the DSS-DM, data on soil mechanics might be integrated into the system. Four categories of soil types are accepted by official governmental codes: Z1, Z2, Z3 and Z4, ranging from massive rock to soft alluvial deposit (Ministry of Public Works and Settlement, Republic of Turkey, 1998). In evaluating the impact of an earthquake on buildings, it might be assumed that it will be higher in a sub-district with a Z3 soil classification than in a sub-district with a Z2 soil categorisation.

The results of the DSS-DM will be used by other sub-systems, particularly for planning the allocation of resources and logistic support. Such models will be designed in the next stage of development of the DSS-DM. The concepts and categories can be updated using inputs from various output user groups, allowing the system to be tailored more to their specific estimation needs. This is considered to be a multidisciplinary and cooperative task, requiring commitment and resources from all parties involved, especially from the central government.

The scenario-based approach is an essential prerequisite for developing adequate preparedness, response and recovery plans, strategies, and organisational structures for disaster management in less-developed contexts where numerous constraints need to be overcome.

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Appendix 1

Residential area

District: Bebek Sub-district: Büyük Bebek					
Cluster name	Number of people during daytime	Number of people during night-time	Type of building	Number of stories	Year of construction
A	175	250	☒ Masonry ☐ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☒ 1–3 ☐ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998
B	600	1,200	☐ Masonry ☒ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☒ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998
C	80	100	☐ Masonry ☐ Reinforced concrete ☒ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998
D	80	100	☐ Masonry ☒ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☒ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998
E	150	200	☐ Masonry ☒ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☐ 4–5 ☒ 6 or more	☐ Before 1940 ☒ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998
F	200	250	☐ Masonry ☒ Reinforced concrete ☐ Wooden masonry composite ☐ Steel Frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☐ 1950–74 ☐ 1975–84 ☒ 1985–97 ☐ After 1998

Elementary and pre-schools

District: Bebek Sub-district: Büyük Bebek					
Name	Number of people during daytime	Number of people during night-time	Type of building	Number of stories	Year of construction
Kindergarten	120	2	☐ Masonry ☒ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☒ After 1998
Tevfik Fikret İlkokulu	350	6	☒ Masonry ☐ Reinforced concrete ☐ Wooden masonry composite ☐ Steel frame	☐ 1–3 ☒ 4–5 ☐ 6 or more	☐ Before 1940 ☐ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998

Universities

District: Bebek Sub-district: Büyük Bebek					
Name	Number of people during daytime	Number of people during night-time	Type of building	Number of stories	Year of construction
Bogaziçi University	3,700	1,150	☐ Masonry ☐ Reinforced concrete ☐ Wooden masonry composite ☒ Steel frame	☐ 1–3 ☐ 4–5 ☐ 6 or more	☐ Before 1940 ☒ 1940–49 ☐ 1950–74 ☐ 1975–84 ☐ 1985–97 ☐ After 1998

Appendix 2*

Table 3 Damage estimations for clusters

Number of people in damaged buildings	District: Bebek Period: day	Earthquake intensity: VIII		
		Collapsed/heavy damage	Moderate damage	Slight damage
Clusters:				No damage
Residential	168	564	1,498	2,111
Elementary and pre-schools	6	19	128	317
Universities	37	37	296	3,330
Sub-districts				
Büyük Bebek	82	235	860	4,278
Cevdet Pasa	55	145	380	320
Çamlı Bahçe	25	60	235	680
Küçük Bebek	27	115	275	308
Asiyan	22	64	172	172
Total in district	211	620	1,922	5,758

Table 4 Damage estimations for construction typologies

Distribution of people in damaged buildings	District: Bebek Period: day	Earthquake intensity: VIII		
		Collapsed/heavy damage	Moderate damage	Slight damage
Buildings by typology:				No damage
Wooden/masonry mixture, 5 stories or less	4	16	28	32
Masonry 1–3 stories	0	7	133	560
Masonry 4–5 stories	0	7	98	245
Steel frame 1940 and after	37	37	296	3,330
Reinforced concrete 5 stories or less 1940–49	0	4	20	56
Reinforced concrete 6 stories or more 1940–49	3	27	45	75
Reinforced concrete 5 stories or less 1950–74	116	350	932	932
Reinforced concrete 6 stories or more 1950–74	20	40	100	40
Reinforced concrete 5 stories or less 1975–84	20	40	100	240
Reinforced concrete 5 stories or less 1985–97	4	80	140	176
Reinforced concrete 5 stories or less 1998 and after	6			72
Total in district	210			5,758

Table 5 Casualty estimations for clusters

Human injuries and deaths	District: Bebek Period: day	Earthquake intensity: VIII		
Clusters:	Deaths	Seriously injured	Moderately injured	Slightly injured
Residential	10	13	20	52
Elementary schools and pre-schools	1	1	1	2
Universities	2	3	4	12
Sub-districts				
Büyük Bebek	5	7	9	26
Cevdet Pasa	3	4	7	17
Çamlı Bahçe	2	2	3	8
Küçük Bebek	2	2	3	9
Asiyan	1	2	2	7
Total in district	13	17	25	66

Table 6 Estimations of need for shelter

Number of people needing shelter Earthquake intensity: VIII		District: Bebek Period: day	
Sub-districts		Number of people	
Büyük Bebek		382	
Cevdet Pasa		797	
Çamlı Bahçe		298	
Küçük Bebek		216	
Asiyan		382	
Total in district		2,075	

* The numbers in the tables are rounded values to report integer casualty estimations.